Tejas Pagare

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Education

Indian Institute of Technology Bombay **Mumbai, India**
Bachelor of Technology in Electrical Engineering and Masters in Control and Computing
Expected Graduation: Jul 2024

Research Experience

Decentralized Learning in Matching Markets *Jan 23 - Present*
Guide: Prof. Avishek Ghosh *Syscon, IIT Bombay*
- Reviewed recent literature on online decentralized learning for two-sided competitive matching markets
- Working on modeling the problem for huge number of agents in the matching markets using clustering framework

Distributed Stochastic Approximation *Sept 22 - Present*
Guide: Prof. Vivek S. Borkar *Electrical Dept., IIT Bombay*
- Analysed the asymptotic behaviour of existing Distributed SA algorithms using **ODE-based analysis**
- Deriving **Concentration Bounds** of Quasi-Linear and Non-Linear Gossip Based Distributed SA algorithms

Exploration in Deep Reinforcement Learning *July 22 - Present*
Guide: Anirudh Goyal and Alex Lamb *Mila, Montréal*
- Reviewed and experimented existing approaches for Deep Exploration in Image Based Deep RL
- Proposed a **Transformer**-based algorithm taking inspiration from Representation Learning and Model Based RL
- Studying the effects of **Perciever IO** Transformer Architecture for Model Based Offline RL

Average Reward Deep Reinforcement Learning *Aug 21 - March 22*
Guide: Prof. Vivek S. Borkar and Dr. Konstantin Avrachenkov *Electrical Dept., IIT Bombay*
- Formulated a Deep RL algorithm based on Full Gradient DQN, **a provably convergent scheme** for **average reward** criterion which outperforms Deep Q-Network (DQN) in terms of convergence rate
- Proposed a variant of Differential Q-Learning with Neural Function Approximation based on Full Gradient DQN
- Extended the algorithm for **restless multi-armed bandits** with average reward in the Whittle framework

[**Paper submitted in an International Control Conference**]

Representation Learning for Robotic Manipulation *Jun 21 - Aug 21*
Guide: Dr. Vibhav Vineet, Senior Researcher *Microsoft Research, Redmond*
- Examined the effects of various Transformer Architectures for Vision (Image and Video Based) and Natural Language for **Imitation** and **Reinforcement Learning** for Robotic Manipulation Tasks
- Experimented Large Pre-trained Image Models for Robotic Manipulation Tasks using **Behavioral Cloning** in PyBullet Simulation and Real Robot at MSR Redmond Lab
- Studied the impact of Transformer-based **Self-Supervised Learning Models** in Reinforcement Learning

Work Experience

Industrial Internship *Mar - May 21*
Daikin Industries Ltd., Guide: Ekant Amin and Ashish Busi *Daikin, Japan*
- Developed a **Key-Value information extraction** pipeline from **Audio** reducing working time by **ten folds**
- Implemented a novel English Accented Speech Recognition (ASR) algorithm based on **Multi-Task learning**
- Applied ASR for Japanese and English voice transcription, including audio longer than 20 seconds
- Added a novel hybrid Heuristic and Neural based data post-processing module for Spelling and Grammar correction

Key Projects

Combinatorial Optimization for Traveling Salesman Problem
Course Project (Optimization in Machine Learning), Guide: Prof. Ganesh Ramakrishnan *Mar 22 - May 22*
- Reviewed and analysed the performance of **Heuristic Based** Methods such as Nearest Neighbour, Near Insertion and Farthest Insertion for the Traveling Salesman Problem using **Concorde** TSP Solver
- Studied RL in the context of Combinatorial Optimization and Implemented Deep RL based on **REINFORCE** with Attention Mechanism and Pointer Network to solve TSP with over 100 nodes.

Speaker Identification
Course Project (Automatic Speech Recognition), Guide: Prof. Preethi Jyothi *Mar 21 - May 21*
- Reviewed some of the recent state-of-the-art Methods for Speaker Identification and Verification
- Implemented the paper **SAASI** (***Interspeech 2020***) which uses Generalized End-to-End (**GE2E**) loss on the embedding obtained by **Self-Attention** and **Adversarial** Loss to make Model robust to Perturbations
- Performed ablation study and achieved a significant increase in performance on **Equal Error Rate** EER evaluation metric for **Unknown Speaker Household** compared to Non-Adversarial Model

Seam Carving for Images
Course Project (Digital Image Processing), Guide: Prof. Ajit Rajwade *Nov 21 - Dec 21*
- Implemented the paper Seam Carving for Content-Aware Image Resizing (***SIGGRAPH'07***) for Forward and Backward Energy models using Dynamic Programming and Graph Cuts in MATLAB
- Performed ablation study and implemented various applications like Object Removal & Content Amplification

Deep Image Prior
Course Project (Machine Learning in Remote Sensing-II), Guide: Prof. Biplab Banerjee *Autumn 20*
- Studied the paper **Deep Image Prior** (***CVPR 2018***) which uses the **randomly-initialized neural network** as a handcrafted prior to obtain excellent results in **standard inverse problems**
- Performed ablation study on Deep Image Prior for Image Denoising, Super-Resolution, and In-Painting tasks

ATM Simulator on Pt-51 Board
Course Project (Microprocessors Laboratory) *Spring 2021*
- Coded a program to simulate an ATM on the **Atmel AT89C51** Micro-controller with a **LCD Module**
- Programmed the micro-controller in **Embedded C** using **Keil μVision** and **Flip** softwares
- Used **UART** Module and **RealTerm** software for interfacing between keyboard and micro-controller

Digital Logic Design in VHDL
Course Project (Digital Systems), Guide: Prof. Virendra Singh *Autumn 2020*
- Designed a **Music Synthesizer** using **VHDL** to play a music automated using **Finite State Machine**
- Designed a signed **16-bit ALU** using **Structural VHDL** which computes addition, subtraction, NAND & XOR
- Performed addition using **16-bit Kogge–Stone Fast Adder** that returns output in 17-bit 2's complement form
- Simulated the circuit using **Quartus** by handpicking test vectors covering all edge cases for each operation

Positions of Responsibility

Team Member | IIT Bombay Mars Rover Team
Image Processing Subdivision Head *Aug 20 - April 22*
Best Navigation Award in the European Rover Challenge, Remote Formula (ERC 2022) among 28 teams from 7 different countries
- Deployed robust **arrow** and **cone** detection algorithm on the Mars rover for successful navigation
- Helped in designing **obstacle avoidance** and **path planning** algorithm in Robot Operating System (ROS)
- Guided Freshman Trainees in the team to familiarise themselves with **Computer Vision** and **Machine Learning**

Hospitality and Assitance Coordinator | Mood Indigo, IIT Bombay
*Asia's largest College Cultural festival, **1,00,000+** viewership, **100+** events, **2000+** colleges* *May - Dec 20*
- Appointing and coordinating with more than **1,000+ College Ambassadors** all across the country
- Organizing weekly events and workshops to engage with **30k+** students from **1000+** colleges
- Assisted in ideating a revamped **Multilevel Publicity Program** to increase outreach of Mood Indigo

Programming Skills

Programming Languages	Python, C++, Julia, Embedded C, Assembly, VHDL
AI/ML Libraries	Pytorch, Tensorflow, RLlib, Numpy, Kaldi, OpenCV, ROS
Tools	Git, LaTeX, AutoCAD, SolidWorks, Intel Quartus Prime

Key Courses Undertaken

Artificial Intelligence	Optimization in Machine Learning, Automatic Speech Recognition, Machine Learning in Remote Sensing-II, Deep Reinforcement Learning
Electrical	Introduction to Stochastic Optimization, Mutlivariable Control System, Applied Linear Algebra, Digital Image Processing, Microprocessors, Communication Systems, Electromagnetic Waves, Power Engineering I & II, Digital Systems, Signal Processing, Probability and Random Processes, Analog Circuits
Mathematics	Differential Equations I & II, Complex Analysis, Linear Algebra, Calculus

Extracurricular Activities

- Successfully completed a year long **National Sports Organization (NSO)** programme in Badminton *(2020)*
- Selected for **Regional Level** Badminton Championship held among Kendriya Vidyalaya's Mumbai Region *(2016)*
- Represented Hostel 2 in Inter-Hostel **Jhatka General Championship** by Electronics Club at IIT Bombay *(2019)*
- **Organizer at Public Relations** department of **E-Summit 2019**, Asia's largest entrepreneurship event held to inspire countless ideas, expedite innovative startups through a cluster of events designed for everyone *(2020)*

References

Prof. Vivek S. Borkar
Electrical Engineering Dept.
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Dr. Vibhav Vineet
MSR Redmond
Vibhav.Vineet@microsoft.com